

Bit Operation Cost of “Holdout” Key-Recovery Attacks Against Classic McEliece

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Abstract. We cost the best Holdout key-recovery attack that we currently know for each Classic McEliece parameter set. One shortened coordinate set of the direct locator-recovery method of Ghoshal, Ishai, Jain, and Sun suffices for every set: public parity-check elimination selects the $mt + 1$ compatible labels needed for key completion and replaces the four or five shortened sets used in the source analysis. We make every sparse linear-system solve reliable by applying Eberly’s random diagonal scaling and iterative scalar Lanczos solver over a larger field, mapping each returned solution back to the original field, and substituting it into the original equations. Separately costing matrix passes that use only binary derivative data and operations with general field coefficients gives complete conditional estimates of $2^{126.77}$ bit operations for `mceliece348864`, $2^{145.22}$ for `mceliece460896`, $2^{137.48}$ for `mceliece6688128`, $2^{136.65}$ for `mceliece6960119`, and $2^{137.48}$ for `mceliece8192128`. The corresponding simultaneously stored state estimates are $2^{51.33}$, $2^{59.10}$, $2^{55.18}$, $2^{54.86}$, and $2^{55.18}$ bits. All five work estimates lie below the NIST classical-gate reference level for their claimed category.

The estimates remain conditional on the source’s conjectures about the canonical form of binary Goppa codes and the rigidity of rank-one solutions, together with the required rank inequalities for the full relation matrices and their coordinate blocks. Failure of either the projected-kernel rank condition or the four-holdout rigidity assumption at sufficiently many coordinates could create an exponential obstruction. The estimates are arithmetic and state estimates, not elapsed-time claims, and no Classic McEliece target key has been recovered.

Note: This is a living document: the estimate, assumptions, and scope will be updated as algorithms and reproducible evidence improve.

Keywords: Classic McEliece · Holdout Attack · Key Recovery · Bit Operations · Sparse Linear Algebra

1 Introduction

Classic McEliece derives its public code from a binary Goppa code [3,9]. Recent Holdout and higher-order-vanishing constructions turn publicly computable polynomial relations into candidate support values, called *locators*, and then

conditionally into an equivalent decoding key [5,6,7,11]. Their concrete cost is sensitive to three details that an algebraic-operation count can hide: how many shortened coordinate sets must be processed, which matrix entries are binary, and whether the algorithm used to solve the sparse linear systems has a quantified failure probability.

Some Terminology. Here a *holdout* is a public coordinate deliberately omitted from a set of vanishing constraints so that the resulting relations can reveal local information at that coordinate. A *projective label* is a proposed locator in the projective line $\mathbb{P}^1(\mathbb{F}_q)$, consisting of the q field elements and one point at infinity. The direct route repeatedly tests such labels at held-out coordinates; accepted labels are then passed to a Goppa-key completion algorithm and full public-key verification.

Current Revision. This paper resolves those three costing questions for the direct locator-recovery route in the 27 August 2026 revision of Ghoshal et al. [5]. We report one best current-route total for each parameter set. The earlier singleton-tangent route (studied in previous revisions of this report) is not retained as a competing model because it costs more while leaving additional attack stages unresolved. That earlier route still requires methods to derive tangent data from the public key, reject incorrect locator guesses, align coordinates that differ by the Frobenius field automorphism $x \mapsto x^2$, and extend a partially recovered support. Necessary conditions in Vedenov’s method for retaining more coordinates after shortening also exclude the small dimensions on which the earlier favorable cost depended. Section 6 records these short rejection arguments without preserving superseded numerical tables.

The calculation is conditional in the precise sense of Section 7. In particular, the paper does not assign probabilities to unproved structural conjectures.

Our contributions:

1. a proof that one shortened coordinate set supplies the $mt + 1$ mutually compatible labels needed by the key-completion algorithm for every Classic McEliece parameter set;
2. a reliable method for solving the affine sparse systems over a larger field, including an exact linear map back to the original field and substitution into the original equations;
3. an explicit decomposition into binary operations and general field operations, with a finite estimate for every remaining algorithmic step; and
4. a small self-contained artifact, `mccost.py`, that recomputes only the dimensions, reliability choices, state estimates, and five current-route totals in this paper.

The public artifact is available at github.com/mjosaarinen/mccost.

2 The Direct Locator-Recovery System

Let m be the extension degree, $q = 2^m$ the field size, n the code length, t the degree of the secret Goppa polynomial, and k the public-code dimension. The National Institute of Standards and Technology (NIST) post-quantum evaluation criteria use 2^{143} , 2^{207} , and 2^{272} classical gates as reference levels for Categories 1, 3, and 5, respectively [10]. These are reference comparisons for the bit-operation metric used here; they are not simply 2^{128} , 2^{192} , and 2^{256} elementary operations.

Table 1 gives the five fixed parameter cells from the source analysis. *Shortening* on ℓ coordinates means restricting to codewords that are zero there and then deleting those coordinates; it gives the remaining length and dimension $(n_\ell, k_\ell) = (n - \ell, k - \ell)$. A polynomial is *homogeneous* when all of its monomials have the same total degree. A Hermite jet at a point is the collection of a polynomial's value and its Hasse derivatives, the coefficient-based derivatives appropriate in finite characteristic, up to a chosen order; multiplicity s means imposing vanishing for derivative orders below s . In the profile column, s^a means that a points receive multiplicity s . We use the source's cells without re-optimizing them.

Table 1. Direct locator-recovery cells and NIST classical-gate reference levels.

Parameter set	Category / reference	(m, n, t, k)	(ℓ, d)	(n_ℓ, k_ℓ)	Profile
mceliece348864	1 / 2^{143}	(12, 3488, 64, 2720)	(2569, 7)	(919, 151)	$5^{42}6^{877}$
mceliece460896	3 / 2^{207}	(13, 4608, 96, 3360)	(3159, 8)	(1449, 201)	$5^46^{177}7^{1268}$
mceliece6688128	5 / 2^{272}	(13, 6688, 128, 5024)	(4802, 7)	(1886, 222)	$5^{35}6^{1851}$
mceliece6960119	5 / 2^{272}	(13, 6960, 119, 5413)	(5198, 7)	(1762, 215)	$5^{24}6^{1738}$
mceliece8192128	5 / 2^{272}	(13, 8192, 128, 6528)	(6306, 7)	(1886, 222)	$5^{35}6^{1851}$

For a detailed derivation, we use mceliece348864 as the running example. Its parameters are

$$(m, n, t, k) = (12, 3488, 64, 2720). \quad (1)$$

Its fixed cell has $(\ell, d) = (2569, 7)$, $(n_\ell, k_\ell) = (919, 151)$, and profile $5^{42}6^{877}$. The same formulas below are evaluated at every row of Table 1 by the artifact.

At a high level, the construction forms two systems of homogeneous polynomial values and derivatives from the shortened binary public generator. For each proposed locator value, rows that fix otherwise arbitrary scaling and rows that test that locator turn the two systems into an affine feasibility test $Ax = b$. The structural premises stated in Section 7 connect accepted solutions to the hidden Goppa locators; the present section only counts the resulting matrices.

For multiplicity s , one homogeneous jet copy has $\binom{k_\ell + s - 1}{s - 1}$ columns. Thus the profile gives

$$M = 42 \binom{155}{4} + 877 \binom{156}{5} = 633,864,594,132. \quad (2)$$

The space of homogeneous degree-seven equations has

$$N_d = \binom{k_\ell + d - 1}{d} = \binom{157}{7} = 407,340,975,756. \quad (3)$$

The four blocks that fix scaling each have length $L_5 = \binom{155}{4} = 23,130,030$. The largest block that tests a proposed locator is

$$B = \binom{152}{2} + \binom{153}{3} + \binom{154}{4} + \binom{155}{5} = 721,656,784. \quad (4)$$

Consequently, the affine feasibility matrix has

$$U = 2M = 1,267,729,188,264, \quad (5)$$

$$R = 2N_d + 4(L_5 - 1) + 1 + B = 815,496,128,413. \quad (6)$$

The two value-and-derivative systems are binary: the shortened public generator and all their zero-one incidence coefficients, which record the relevant monomial-derivative containments, lie in the two-element field $\mathbb{F}_2$. A direct count from the multiplicity profile gives

$$z_{\text{bin}} = 2N_d \left(42 \sum_{r=0}^4 \binom{7}{r} + 877 \sum_{r=0}^5 \binom{7}{r} \right) = 89,124,576,131,509,776 \quad (7)$$

nonzero binary entries. Only the scaling-fixing and locator-testing pieces require arbitrary coefficients from the base field $\mathbb{F}_{2^{12}}$; their two copies have

$$z_{\text{gen}} = 2(4(L_5 - 1) + 1 + B) = 1,628,353,802 \quad (8)$$

nonzero entries. Treating both populations as general field multiplications would therefore discard the main circuit-level distinction.

Choose three coordinates of multiplicity five as reference hold-outs and prescribe three distinct projective rows. A projective linear transformation, represented by an invertible two-by-two matrix up to nonzero scale, can carry any ordered triple of distinct projective points to those rows. Under Assumption 1, the preliminary three-holdout scan at a fourth coordinate accepts exactly the m normalized Frobenius-conjugate branches. Any one of them may be retained because it fixes one globally compatible conjugate rather than an independent local choice. Appending that fourth row leaves exactly one accepted label at every coordinate in the four-holdout scan. The preliminary scan uses $q + 1$ feasibility systems and the full scan uses $n_\ell(q + 1)$; the preliminary systems are smaller, so we conservatively charge all of them at the largest four-holdout size:

$$S = (n_\ell + 1)(q + 1) = 3,769,240. \quad (9)$$

3 Why One Shortened Chart Suffices

The direct construction labels every position of one shortened coordinate set. Its shortening gate verifies that the deleted public columns have full rank $|I|$, so

the shortened code has dimension $k - |I|$. The source analysis covers the entire public support with four or five such sets, but the known-point key-completion algorithm needs only $mt + 1$ labels expressed in one common projective coordinate system [8]. The following public reduction supplies them in every row of Table 1.

Lemma 1 (One-chart selection). *Let I be the shortening set, let $J = [n] \setminus I$ be the remaining positions, and let H be a full binary parity-check matrix, so the public code is $\ker H$. If the shortened systematic construction has dimension $k - |I|$ and $|J| \geq n - k + 1$, then $\text{rank } H_J = n - k = mt$. Hence the recovered labels on J contain mt positions whose parity-check columns are linearly independent and one further labeled position.*

Proof. Extending a shortened word by inserting zeros at I identifies $\ker H_J$ with the public code shortened on I . This kernel has dimension $k - |I|$. The rank-nullity theorem gives

$$\text{rank } H_J = |J| - (k - |I|) = n - k = mt.$$

Gaussian elimination over $\mathbb{F}_2$ publicly selects mt independent columns. The inequality $|J| \geq mt + 1$ leaves at least one further labelled position.

Table 2. Capacity of one shortened chart for known-point completion.

Parameter set	n_ℓ	mt Required	$mt + 1$ Surplus
mceliece348864	919	768	769 150
mceliece460896	1,449	1,248	1,249 200
mceliece6688128	1,886	1,664	1,665 221
mceliece6960119	1,762	1,547	1,548 214
mceliece8192128	1,886	1,664	1,665 221

Lemma 2 (Projective-coordinate completion). *Suppose the labels recovered on one shortened chart have the form $z_i = \phi(\alpha_i^{2^e})$, where the $\alpha_i \in \mathbb{F}_{2^m}$ are distinct hidden locators, e is one common exponent, and ϕ is one common projective linear transformation. Enumerating the projective values absent from any selected $mt + 1$ labels includes a value that converts those labels into an affine transform of the same Frobenius conjugate. On that trial the known-point key-completion algorithm receives a valid equivalent binary Goppa representation.*

Proof. Let $p = \phi(\infty)$. Every hidden locator is finite, so p is absent from every z_i and in particular from the selected labels. Choose a projective transformation ψ_p that sends p to infinity. The composition $\psi_p \circ \phi$ fixes infinity and is therefore affine: for some $a \in \mathbb{F}_{2^m}^\times$ and $b \in \mathbb{F}_{2^m}$, the transformed labels are $\beta_i = a\alpha_i^{2^e} + b$.

Write $g^{(2^e)}$ for the polynomial obtained by applying $x \mapsto x^{2^e}$ to the coefficients of the hidden monic degree- t Goppa polynomial g . Because binary code-word coordinates are fixed by this field automorphism, applying it coefficientwise to the Goppa syndrome criterion and then substituting $Y = aX + b$ gives

$$\Gamma(\alpha, g) = \Gamma(\alpha^{2^e}, g^{(2^e)}) = \Gamma(\beta, g_p), \quad g_p(Y) = a^t g^{(2^e)}\left(\frac{Y - b}{a}\right).$$

Thus the correct pole produces indexed affine labels for an equivalent degree- t binary Goppa key. The deterministic completion theorem of Kirshanova and May applies after Lemma 1 selects more than mt labels whose public parity-check columns have rank mt [8, Theorem 4]. Full public-key verification accepts a valid equivalent key and rejects an invalid output.

To apply the affine recovery algorithm, one projective value must be designated as the point at infinity, or *pole*. For each projective value absent from the selected labels, the completion step chooses it as the pole, converts the other $mt + 1$ labels to affine field values, applies the degree- t Goppa-polynomial recovery algorithm, extends the candidate support, and accepts only a complete public-key verification. By Lemma 2, one trial necessarily chooses a compatible pole. We charge every possible pole in the deliberately loose finite estimate of Section 5; this completion work is smaller than the locator scan in every row. Thus one chart uses $(n_\ell + 1)(q + 1)$ largest-system equivalents, with no multiplier for processing several shortened coordinate sets.

4 A Reliable Affine Sparse Solve

Write $\mathbb{F}_{2^m}$ for the field used by the public construction. Each candidate asks whether the affine linear system

$$Ax = b, \quad A \in \mathbb{F}_{2^m}^{R \times U}, \quad b \in \mathbb{F}_{2^m}^R, \quad (10)$$

is consistent, meaning that some x satisfies every equation. The matrix is sparse because almost all of its entries are zero. A heuristic solver is inadequate for a headline tally: millions of trials amplify even a small probability of failing on a solvable system. We instead use Eberly's reliable scalar Lanczos construction [4]. Scalar Lanczos is an iterative linear-system algorithm that accesses a matrix through products with vectors. Eberly first multiplies rows and columns by random nonzero diagonal entries; this *preconditioning* makes the required rank and solver properties fail only with a bounded probability.

Choose δ divisible by m and write $\mathbb{F}_{2^\delta}$ for the resulting field $\mathbb{F}_{2^\delta}$. Then $\mathbb{F}_{2^\delta}$ contains a unique subfield of size 2^m , which we identify with $\mathbb{F}_{2^m}$. Sample independent diagonal matrices $D_\alpha \in \mathbb{F}_{2^\delta}^{U \times U}$ and $D_\beta \in \mathbb{F}_{2^\delta}^{R \times R}$ with uniformly random nonzero diagonal entries, and sample a uniform vector $z \in \mathbb{F}_{2^\delta}^U$. Define

$$G = D_\alpha A^T D_\beta^2 A D_\alpha \quad (11)$$

and apply scalar Lanczos to

$$Gy = D_\alpha A^T D_\beta^2 (b - Az). \quad (12)$$

Return $x = z + D_\alpha y$. A *retraction* is an $\mathbb{F}_{2^m}$ -linear map from $\mathbb{F}_{2^\delta}$ back to $\mathbb{F}_{2^m}$ that fixes every element of $\mathbb{F}_{2^m}$; apply it to every coordinate of x . Finally, perform *exact replay*: substitute the retracted vector into the original equation (10) and refuse it unless equality holds.

Theorem 1 (Reliable affine reduction). *If (10) is consistent, the right-hand side of (12) is the image under G of a uniformly random vector, which is the input distribution required by Eberly’s scalar-Lanczos theorem. When the random diagonal scaling preserves the required rank and Lanczos succeeds, the returned vector solves (10). An inconsistent or corrupted output cannot be accepted by exact replay. Moreover, a solution over $\mathbb{F}_{2^\delta}$ can be mapped to a solution over $\mathbb{F}_{2^m}$ by a fixed $\mathbb{F}_{2^m}$ -linear retraction.*

Proof. Fix one solution x_0 of (10). Its randomized right-hand side is

$$D_\alpha A^T D_\beta^2 (b - Az) = G D_\alpha^{-1} (x_0 - z).$$

Since z is uniform, the hidden preimage is uniform in $\mathbb{F}_{2^\delta}^U$. This is exactly the image-of-a-random-vector distribution in Eberly’s theorem. When $\text{rank } G = \text{rank } A$, the kernels of G and AD_α agree. Hence any solution y of (12) satisfies $A(z + D_\alpha y) = b$.

Because $m \mid \delta$, $\mathbb{F}_{2^\delta}$ is an extension of $\mathbb{F}_{2^m}$. Choose an $\mathbb{F}_{2^m}$ -linear map $\pi : \mathbb{F}_{2^\delta} \rightarrow \mathbb{F}_{2^m}$ with $\pi(1) = 1$, and apply it coordinatewise. It fixes $\mathbb{F}_{2^m}$, so $A\pi(x) = \pi(Ax) = \pi(b) = b$. Finally, exact replay rejects every vector that does not satisfy the original system, regardless of how it arose.

For a matrix with at most U columns and rank at most R , Eberly’s bounds give, per system,

$$p_{\text{pre}} \leq \frac{11U^2 - U}{2(2^\delta - 1)}, \quad p_{\text{L}} \leq \frac{R(R+1)}{2^\delta}. \quad (13)$$

Here p_{pre} bounds failure of the random diagonal scaling and p_{L} bounds failure of scalar Lanczos. Adding these probabilities for every one-chart system gives a valid upper bound even when the failure events are not independent. For each parameter set, the artifact selects the least-work multiple of m satisfying $S(p_{\text{pre}} + p_{\text{L}}) < 2^{-128}$. For the running example this gives

$$\delta = 240, \quad \log_2(S(p_{\text{pre}} + p_{\text{L}})) < -135.1793. \quad (14)$$

This is a bound on algorithmic solver failure. It does not measure the truth of the structural assumptions in Section 7.

5 Bit-Operation Tally

Represent each element of $\mathbb{F}_{2^\delta}$ by the δ binary coefficients of a polynomial. Our direct, or schoolbook, multiplication estimate charges δ^2 Boolean AND operations, $\delta^2 - (2\delta - 1)$ exclusive-or (XOR) operations to combine equal-degree products, and $\delta(\delta - 1)$ XOR operations to reduce the result modulo the fixed irreducible polynomial that defines the field. We therefore charge $C_\times$ for one multiplication and C_{ma} for one multiplication followed by an addition:

$$C_\times(\delta) = 3\delta^2 - 3\delta + 1, \quad C_{\text{ma}}(\delta) = 3\delta^2 - 2\delta + 1. \quad (15)$$

At $\delta = 240$, these are 172,081 and 172,321 bit operations. This transparent circuit estimate does not use an assumed multiplier for an unspecified field implementation.

One multiplication of a vector by the preconditioned matrix G in (11) is charged by

$$C_G = 2z_{\text{bin}}\delta + (2z_{\text{gen}} + 2U + R)C_{\text{ma}}(\delta) = 43,357,797,573,481,659,425. \quad (16)$$

The first term consists only of XOR operations on bit vectors. The second conservatively charges the arbitrary field coefficients and every diagonal scaling as a full multiply-add. Between matrix products, scalar Lanczos updates a fixed collection of vectors and inner products; we estimate one such recurrence step and the combined initialization and finalization as

$$C_{\text{rec}} = (17U + 4\delta + 3)C_{\text{ma}}(\delta), \quad (17)$$

$$C_{\text{init}} = (20U + 4\delta + 3)C_{\text{ma}}(\delta). \quad (18)$$

The estimated cost of substituting one candidate into the original base-field equations is

$$C_{\text{replay}} = z_{\text{bin}}m + z_{\text{gen}}C_{\text{ma}}(m) + Rm. \quad (19)$$

The estimated cost of generating the random diagonal entries and affine shift, then applying a general m -by- δ binary linear map to every output coordinate, is

$$C_{\text{source}} = (2U + R + mU)\delta. \quad (20)$$

These costs are included in each feasibility-system row of Table 3. Charging $R+3$ multiplications by G and R recurrence steps gives

$$C_{\text{sys}} = (R + 3)C_G + RC_{\text{rec}} + C_{\text{init}} + C_{\text{replay}} + C_{\text{source}} \approx 2^{104.92}. \quad (21)$$

The public one-chart selection estimate for Gaussian elimination is

$$C_{\text{select}} = n_\ell(n - k)^2 + n_\ell(n - k) \approx 2^{29.02}. \quad (22)$$

Put $r = mt = 768$ for the number of parity-check rows and $L = 769$ for the number of selected labels. Enumerating every possible missing pole and, conservatively, permitting even a field inversion for every algebraic operation gives the finite estimate

$$C_{\text{finish}} = (q + 1 - L)(qL^6 + qnr^4) 2mC_\times(m) \approx 2^{94.45}. \quad (23)$$

The key-completion work is well below the locator scan and changes no reported headline digit. The complete conditional estimate is therefore

$$C_{\text{total}} = SC_{\text{sys}} + C_{\text{select}} + C_{\text{finish}} \approx 2^{126.77} \text{ bit operations.} \quad (24)$$

Table 3. Detailed conditional attack tally for `mceliece348864`.

Item	Bit-operation estimate
One affine feasibility system	$2^{104.92}$
All 3,769,240 systems	$2^{126.77}$
Public label selection	$2^{29.02}$
Projective completion and verification	$2^{94.45}$
Complete conditional tally	$2^{126.77}$

Applying the same formulas to every fixed cell gives Table 4. Work is measured in bit operations and state in simultaneously stored bits. The solver-failure column is the union bound over every charged affine system; the margin subtracts the work exponent from the NIST classical-gate reference exponent for the claimed category.

Table 4. Best known conditional Holdout attack tally for every Classic McEliece parameter set.

Parameter set	Reference	Solver	Work	State	Failure	Margin
<code>mceliece348864</code>	1 / 2^{143}	$\mathbb{F}_{2^{240}}$	$2^{126.77}$	$2^{51.33}$	$< 2^{-135.17}$	16.23
<code>mceliece460896</code>	3 / 2^{207}	$\mathbb{F}_{2^{260}}$	$2^{145.22}$	$2^{59.10}$	$< 2^{-138.19}$	61.78
<code>mceliece6688128</code>	5 / 2^{272}	$\mathbb{F}_{2^{247}}$	$2^{137.48}$	$2^{55.18}$	$< 2^{-132.51}$	134.52
<code>mceliece6960119</code>	5 / 2^{272}	$\mathbb{F}_{2^{247}}$	$2^{136.65}$	$2^{54.86}$	$< 2^{-133.25}$	135.35
<code>mceliece8192128</code>	5 / 2^{272}	$\mathbb{F}_{2^{247}}$	$2^{137.48}$	$2^{55.18}$	$< 2^{-132.51}$	134.52

The *active state* is the information that must be stored simultaneously. For each row the estimate retains eight vectors of length U for the recurrence and random shift, two work vectors of length R , and the z_{gen} arbitrary field coefficients. For the running example,

$$C_{\text{state}} = (8U + 2R + z_{\text{gen}})\delta \approx 2^{51.33} \text{ bits.} \quad (25)$$

This is a live-state estimate for the mathematical schedule, not a claim that the computation fits a particular machine. Memory traffic, communication, check-pointing, and redundant execution are not converted into additional bit operations.

6 Why Other Proposed Savings Do Not Change the Tally

Earlier one-coordinate route. Previous revisions priced a route that built a separate relation space after omitting each one of several selected coordinates. It then attempted to recover derivative information at those coordinates, guess some locator values, align the resulting values across Frobenius automorphisms, and interpolate the hidden curve. Its lowest numerical subtotal was larger than Table 3 and omitted more unresolved work: constructing the required public relation spaces, rejecting an incorrect tuple of locator guesses, extending the correct partial support, and aligning coordinates were still conditional. The 30 August 2026 revision of Vedenov’s method for retaining additional curve coordinates after shortening also requires $k_\ell \geq 620$ at its first parameter choice satisfying the necessary inequalities, excluding the small shortened dimensions on which that route’s favorable cost depended [11]. It therefore cannot improve the present headline, and we do not reproduce the obsolete scan or tables. Apon’s algebraic-geometric obstruction further explains why a few guessed locators alone cannot identify the binary-Goppa curve [1].

Reconstructing only part of a Wiedemann solution. The Wiedemann method solves sparse linear systems by generating sequences of matrix–vector products. Chou and Niederhagen optimize its use in extended linearization (XL) by reconstructing only the selected coordinates that encode their candidate solution [2]. The direct Holdout route instead needs every coordinate of a relation vector to check its derivatives and continue key recovery, so requesting all coordinates restores a full reconstruction pass. Choosing different numbers of vectors on the left and right can reduce one constant in the sparse matrix passes, but no complete-output reliability and state analysis is currently available for this matrix shape. We therefore make no change to Table 3.

7 What the Estimate Does and Does Not Establish

The following parts of all five tallies in Table 4 are proved, explicitly estimated, or, for randomized solver failure, given finite probability bounds in this paper:

1. the no-slack multiplicity identity, $d > s$, the power-of-two condition on $s - 1$, four available multiplicity-five hold-outs, the exact fixed dimensions, and the separation of nonzero entries into binary and arbitrary-field entries;
2. the rank- mt one-chart selection lemma and the $mt + 1$ label-capacity check for every parameter set;
3. the randomized conversion of each affine system to the form required by Eberly’s reliable solver theorem;
4. the linear map back to the base field, exact substitution into the original system, and the aggregate solver-failure bound;
5. the projective-to-affine key reduction, exhaustive pole estimate, deterministic key-completion preconditions, polynomial-basis circuit costs, and active-state estimates.

The route remains conditional on:

Assumption 1 (Imported structural premises) *For each cell in Table 1, let $E_\ell \in \mathbb{F}_{2^m}^{N_{d,\ell} \times M_\ell}$ be the public value-and-derivative matrix of the shortened binary Goppa instance before the hold-out normalization and locator test. Let B_j be the coordinates of the derivative block used to test the locator at position j , and let π_j project a kernel vector onto that block. We assume the following finite-instance consequences of the canonical-form and rank-one-rigidity conjectures, together with the associated rank predictions of Ghoshal et al. [5, Conjectures C.2 and C.3 and Sec. 7.4]:*

1. Canonical rank-two branch. *For each three- or four-holdout system used by the scan, there are normalized relation vectors $c_\tau^* \in \ker(E_\ell)$ whose stack C^* has rank two and factors as $C^* = U^*V^*$. Writing α_j for the hidden locator, the row of U^* belonging to hold-out τ is $A_\tau(\alpha_\tau^4, 1)$ for some nonzero $A_\tau \in \mathbb{F}_{2^m}$, and the two rows of the right factor obey $(V_2^*)|_{B_j} = \alpha_j^4(V_1^*)|_{B_j}$ on every locator block. Each relation has value 1 and all its positive-order derivative coordinates equal to zero at its own hold-out. Consequently, the true normalized support gives a solution of every affine system that the attack expects to be consistent.*
2. Preliminary Frobenius branches. *With three prescribed projective rows, the locator test at the fourth hold-out accepts exactly the m normalized Frobenius conjugates predicted by the source. Choosing any one of them fixes one exponent $e \in \{0, \dots, m-1\}$ and a compatible fourth row; it does not select an independent value at each coordinate.*
3. Four-holdout rank-one rigidity. *For the resulting four-row factor U^* , there are one exponent e and one projective linear transformation ϕ such that the feasibility systems at every surviving position j accept exactly the single projective label $z_j = \phi(\alpha_j^{2^e})$, with the degenerate outcome representing the point at infinity when necessary. In particular, additional local kernel relations do not create a second accepted label. Because $x \mapsto x^4$ is a field automorphism in characteristic two, the accepted fourth power determines its label uniquely.*
4. Rank admission. *The actual full and block-projected kernels satisfy*

$$\text{rank}(E_\ell) < N_{d,\ell}, \quad \dim \pi_j \ker(E_\ell) < |B_j| \quad \text{for every surviving position } j.$$

These are the two rank conditions required by the conditional recovery theorem. The source's full-matrix and block-deletion rank predictions are used to select the cells in Table 1; this paper assumes that the displayed cells satisfy the two inequalities, but does not need equality with every predicted nullity.

The four premises have different failure modes. Failure of the canonical branch can remove the true solution, so the attack may refuse rather than incur a hidden finite cost. Failure of the preliminary three-holdout statement can leave inequivalent fourth-row choices; trying every possible preliminary survivor would add as much as one factor of $q+1$, which is not included in the tally. Together, four-holdout rigidity and the local projected-kernel inequality exclude the main

known exponential branch obstruction. Let $\mathcal{C}_j$ be the set of labels accepted at coordinate j , and let $\mathcal{T}$ be the family of $(mt + 1)$ -element subsets $T \subseteq J$ whose public parity-check columns have rank mt . A direct enumeration fallback can choose the most favorable such set, but its candidate-tuple search space still has size

$$N_{\text{direct}} = \min_{T \in \mathcal{T}} \prod_{j \in T} |\mathcal{C}_j|.$$

If every admissible T contains at least r coordinates with two or more surviving labels, then $N_{\text{direct}} \geq 2^r$, which is exponential when $r = \Omega(n_\ell)$. This is a possible obstruction for direct enumeration, not a lower bound on all reconciliation strategies: non-singleton candidates may be globally correlated and admit a more efficient reconciliation. Failure of the local projected-kernel inequality can make a coordinate block uninformative and can feed the same obstruction; failure of the full-matrix inequality removes the source theorem’s admission guarantee. The source predicts margins of at least 230 dimensions for the full-matrix inequality and at least 223 dimensions for every projected-block inequality in the displayed cells, which is evidence but not a proof [5, Sec. 7.4, Eq. (11)].

Under Assumption 1, none of those branch multipliers remains: the preliminary choices are the m equivalent Frobenius conjugates, every four-holdout locator set is a singleton in one common coordinate system, at most $q + 1 - (mt + 1)$ missing poles are tried, and the known-point completion takes deterministic polynomial time. The sparse solver has only the separately bounded probability of missing a consistent system and cannot accept an inconsistent one after exact replay. Thus Assumption 1 contains every presently known potential exponential obstruction in this route; this statement does not prove the assumptions or rule out an undiscovered obstruction in their underlying algebra. No probability is assigned to them. No target-scale implementation of the full locator scan has produced public relations, locators, a Goppa polynomial, or a Classic McEliece key. Subject to that boundary, the best attacks we currently know how to tally lie below the corresponding NIST classical-gate reference level for all five Classic McEliece parameter sets.

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Use of Artificial Intelligence

These results originate from a systematic effort to implement and improve crypt-analytic attacks on Post-Quantum Cryptography with AI help. We used OpenAI GPT-5.6-Sol and Claude Opus 5 for theory formulation, proofs, analysis, and preparing and executing experiments. Furthermore, we used the LLMs and Grammarly to prepare and improve the presentation.

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A Artifact Reproduction

Run

```
python3 mccost.py
```

with Python 3. This command recomputes all five current-route reports and checks their exact arithmetic. The artifact intentionally contains no superseded parameter scans or speculative experiments.

B Changelog

2026-08-24. Initial release, followed by same-day cleanup and clarification.

2026-08-27. Updated the Introduction and asymptotic discussion for the then-current revisions by Vedenev and by Ghoshal et al.; added the proof for the rank of one point’s derivative-row block, exact padding of rectangular systems, active-state accounting, joint parameter optimization, and normalization by projective linear transformations. The artifact then separately reconstructed the published conditional competing baseline without importing its unitemized optimized claim.

2026-08-29. Updated for Vedenev’s revision dated 2026-08-28: adopted its rank test for rejecting incorrect guesses and the corresponding sensitivity based on the probability that a random matrix loses rank. Selected parameter choices and attack subtotals were unchanged.

2026-08-30. Sections 2–5 now give the one-chart direct locator-recovery route, reliable larger-field solves followed by exact maps to the original fields, and one best current estimate for each of the five parameter sets, compared with the cited NIST classical-gate reference levels. Superseded singleton tables and artifact functions were removed; Section 6 reflects Vedenev’s current feasibility boundary, and Section 7 records the revised scope and remaining assumptions. We also updated and expanded Assumption 1 later that day.